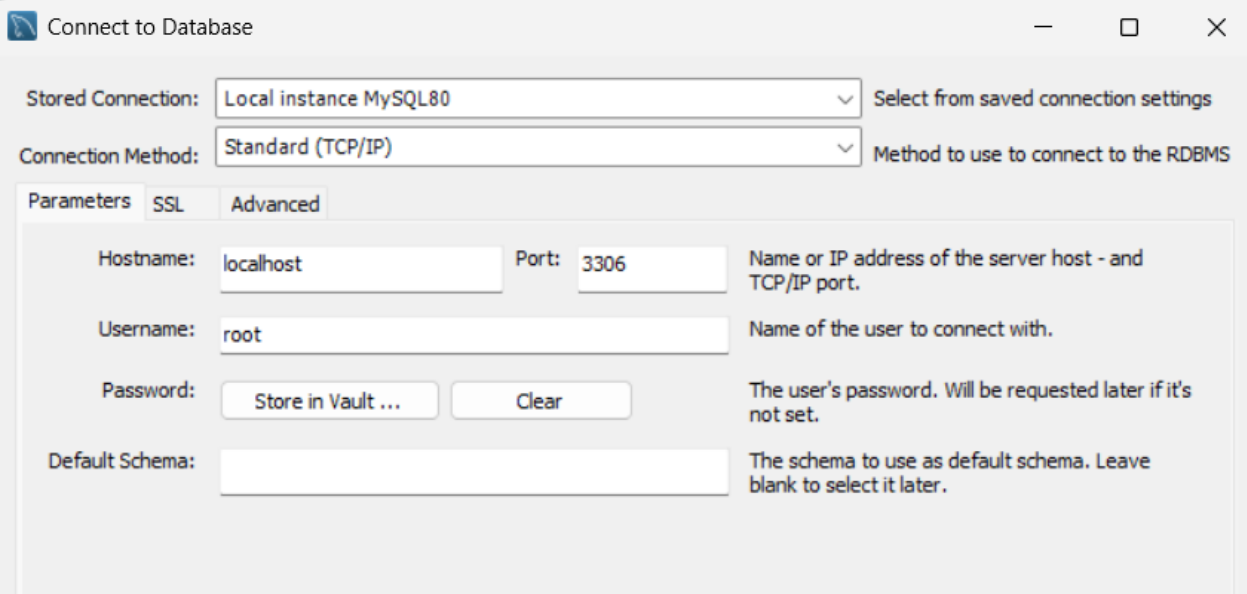
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Java Server/Client Project

This project demonstrates CRUD functions, database management, and Java programming skills. This project is a character creation database that allows the user to utilize CRUD functions to manipulate data.

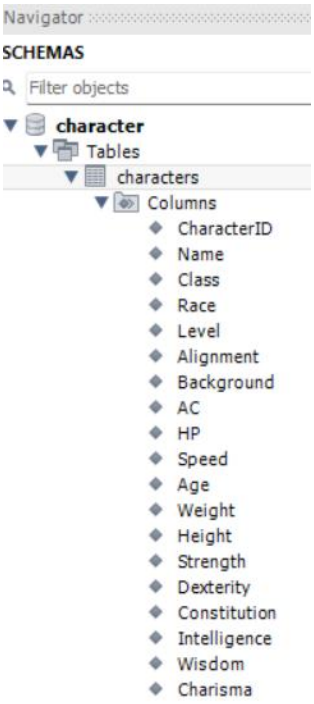
Brief instructions on how to create your own project:

1. We used IntelliJ Idea for this project.
2. The settings for this project should be able to run multiple configurations to mimic a server/client relationship. You can edit this. Ensure you are in the correct directory.
3. You will have to set up mySQL workbench onto your system.
4. Once installed, under the database tab, select "connect to database"
5. Make sure you note the port number, Username should be root, localhost should be listed for the Hostname (we are running this on the local system for this demonstration) You can store your own personal password in the vault.



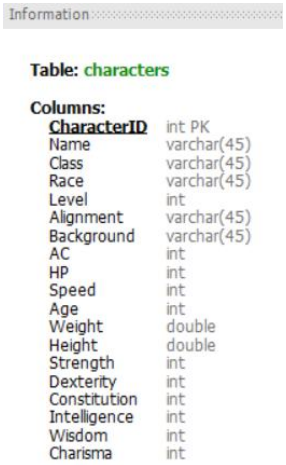
6. This database is named "character"
7. The table is named "characters"

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8. Columns & associated Datatype:

1) CharacterID int PK
2) Name varchar(45)
3) Class varchar(45)
4) Race varchar(45)
5) Level int
6) Alignment varchar(45)
7) Background varchar(45)
8) AC int
9) HP int
10) Speed int
11) Age int
12) Weight double
13) Height double
14) Strength int
15) Dexterity int
16) Constitution int
17) Intelligence int
18) Wisdom int
19) Charisma int



9. If you right click “characters” table and select “Select Rows – Limit 1000” if will show a result grid with each of the characters listed.
10. Open the folder in IntelliJ, and ensure that there are the following files present:
    1) Main.java
    2) Database.java
    3) Server.java
    4) Client.java
    5) CreateCharacter.java
11. In the Database.java folder, edit the URL to reflect the correct information.
    1) String url = “jdbc:mysql://HOSTNAME:PORT/TABLE”;
    2) Fill in the hostname, port, and table name information
    3) See image for reference

```
//REQUIRED COMPONENTS
//need URL, username, and password
String url = "jdbc:mysql://localhost:3306/character"; 22 usages
//normally you would need IP address for the system, but we are using our local system
String username = "root";  22 usages
```

12. In the Database.java file, your password must reflect the password used when connecting mysql
13. Configure IntelliJ to run multiple instances so that you can have multiple clients connected to the server at once.
14. When you right-click the table name in mySQL you will be able to “Select Rows – limit 1000” and it will display the data in the database. Refreshing this will allow you to see the manipulated data after changes are made by the client.

